

5COSC022C.2 Client Server Architecture (2024) - Semester 2 - Coursework - 28th April 2025				
Marker		UoW ID		Student Name
Cassim Farook		w2053223		MuhammedMazhar
Total			83.5	
Model Class Implementation	Mark	Given		
Book Class	2	1	Correct attributes, getters, setters (1) One constructor that initializes some of the attributes. (1) primitive data types found (-1), Using string instead of numeric for price etc (-0.5)	
Author Class	2	2	Correct attributes, getters, setters (1) One constructor that initializes some of the attributes. (1) primitive data types found (-1), Using string instead of numeric (-0.5)	
Customer Class	2	2	Correct attributes, getters, setters (1) One constructor that initializes some of the attributes. (1) primitive data types found (-1), Using string instead of numeric (-0.5)	
Cart Class (to support CartResource)*	1	1	Correct attributes, getters, setters (0.5) At least the default constructor (0.5) primitive data types found (-0.5), Using string instead of numeric for qty etc (-0.5)	
Order Class (to support OrderResource)*	1	0.5	Correct attributes, getters, setters (0.5) At least the default constructor (0.5) primitive data types found (-0.5), Using string instead of numeric for qty etc (-0.5)	
	8	6.5		
Distinct DAO Implementation			*these operations may be written in the resource class, the marks is for students who implemented a DAO pattern given in the lecture notes.	
BookDAO Class (1 Marks)	1	0.75	All CRUD methods For update and delete to indicate they were successful. Can return a boolean, object or throw exception to indicate the status (-0.5) DAO suffix naming convention not used reduce (-0.25)	
AuthorDAO Class (1 Marks)	1	0.75	All CRUD methods For update and delete to indicate they were successful. Can return a boolean, object or throw exception to indicate the status (-0.5) DAO suffix naming convention not used reduce (-0.25)	
CustomerDAO Class (1 Marks)	1	0.75	All CRUD methods For update and delete to indicate they were successful. Can return a boolean, object or throw exception to indicate the status (-0.5) DAO suffix naming convention not used reduce (-0.25)	
CartDAO Class (1 Marks)	1	0.75	All CRUD methods For update and delete to indicate they were successful. Can return a boolean, object or throw exception to indicate the status (-0.5) DAO suffix naming convention not used reduce (-0.25)	
OrderDAO Class (1 Marks)	1	0.25	All CRUD methods For update and delete to indicate they were successful. Can return a boolean, object or throw exception to indicate the status (-0.5) DAO suffix naming convention not used reduce (-0.25)	
	5	3.25	Repository classes could be named as DAO	
Restful Resource Implementation				
Create		@POST and Media Type to be JSON. HTTP status code 201 is required. @Consumes is required. If returning resource object in the payload the @Produces is needed. This can be at class level		Update @PUT Media Type to be JSON. HTTP status code 204 is default. The @Consumes is required. If returning resource object in the payload the @Produces is needed with status code 200. This can be at class level. If method fails the status code cannot be 2x.

Read			@GET and Media Type to be JSON. HTTP status code 200 or 404 (if no data). @Produces is required. This can be at class level. Student to be tested if he has put a if condition to check this logic in the get by id method.	Delete		@DELETE Media Type to be JSON. HTTP status code 204 is required. @Consumes needed to enforces content type JSON header check, this can be set at class level. Should not return a resource object, but can return a message object with 204. If method fails the status code cannot be 2x.						
BookResource Class	5	4.25	Includes all HTTP requests and methods that are in-line with CRUD operations. Appropriate endpoints are used for each method.(1x4).	3.75	Create		Read		Update		Delete	
			Logs provide detailed information suitable for debugging (0.25).	0.00	@POST	0.25	@GET	0.25	@PUT	0.25	@DELETE	0.25
				0.25	Media Type to be specified.	0.25	Media Type to be	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25
			Implementing and using BookNotFoundException (0.25).	0.25	HTTP status code 201.	0.25	code 200 if data	0.25	HTTP status code 204/ 200 if data	0.25	HTTP status code 204.	0.25
			The response should include information about the Author associated with the book. If only the author id, no marks (0.25).	0.00	@Consumes	0.25	@Produces	0.25	@Consumes	0.25	@Consumes	0.25
			Data Validation: Input data is properly validated for POST and PUT (0.25)	0.25								
			Storage Penalty : The model class need to be filled and put in a arraylist or suitable variable. Student can do this in the resource class or in the DAO class. (-0.5)	0.00	return data @Produces missing	0.00	Get all method and		return data @Produces missing	0.00	return data @Produces missing	0.00
							no results 404 missing	-0.25	invalid item or failed with 200	0.00	invalid item or failed with 200	0.00
BookResource Demo * 1 mark moved to using a DAO	9	8.50	All endpoints are demonstrated clearly in Postman (5)	4.50	GET /books			0.50	get all need to return 404 if empty list (-0.50)			
			Including various valid and invalid inputs and expected outputs/errors	1.00	GET /books/{id}			1.00				
			Shows effective use of HTTP methods and status codes (200,201,204,404)	1.00	POST /books			1.00				
			Covers positive test cases Creating a resource with valid data, getting a resource that exists	1.00	PUT /books/{id}			1.00				
			Covers negative test cases Creating resource with missing fields,ists, getting a resource that does not exist	1.00	DELETE /books/{id}			1.00				
AuthorResource Class	5	4.50	Includes all HTTP requests and methods that are in-line with CRUD operations. Appropriate endpoints are used for each method.(1x4).	3.75	Create		Read		Update		Delete	
			Logs provide detailed information suitable for debugging (0.25).	0.00	@POST	0.25	@GET	0.25	@PUT	0.25	@DELETE	0.25
				0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25
			Implementing and using AuthorNotFoundException (0.25).	0.25	HTTP status code 201.	0.25	code 200 if data	0.25	HTTP status code 204/ 200 if data	0.25	HTTP status code 204.	0.25
			Implementing GET /authors/{id}/books to get books by author (0.25).	0.25	@Consumes	0.25	@Produces	0.25	@Consumes	0.25	@Consumes	0.25
			Data Validation: Input data is properly validated for POST and PUT (0.25)	0.25								
			Storage Penalty : The model class need to be filled and put in a arraylist or suitable variable. Student can do this in the resource class or in the DAO class. (-0.5)	0.00	return data @Produces missing	0.00	Get all method and getby id if		return data @Produces missing	0.00	return data @Produces missing	0.00
							no results 404 missing	-0.25	invalid item or failed with 200	0.00	invalid item or failed with 200	0.00
AuthorResource Demo * 1 mark moved to using a DAO	9	8.00	All endpoints are demonstrated clearly in Postman (5)	4.00	GET /authors			0.50	GET /authors/{id}/books			0.50
			Including various valid and invalid inputs and expected outputs/errors	1.00	GET /authors/{id}			1.00	get all need to return 404 if empty list (-0.50)			

			Shows effective use of HTTP methods and status codes (200,201,204,404)	1.00	POST /authors			1.00				
			Covers positive test cases		PUT /authors/{id} (0.5)			0.50				
			Creating a resource with valid data, getting a resource that exists	1.00								
			Covers negative test cases		DELETE /authors/{id} (0.5)			0.50				
			Creating resource with missing fields, getting a resource that does not exist	1.00								
CustomerResource Class	5	4.50	Includes all HTTP requests and methods that are in-line with CRUD operations. Appropriate endpoints are used for each method.(1x4).	3.75	Create		Read		Update		Delete	
			Logs provide detailed information suitable for debugging (0.25).	0.00	@POST	0.25	@GET	0.25	@PUT	0.25	@DELETE	0.25
					Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25
			Implementing and using CustomerNotFoundException (0.25).	0.25								
			The response should include information about the Customer (0.25).	0.25	HTTP status code 201.	0.25	code 200 if data	0.25	HTTP status code 204/ 200 if data	0.25	HTTP status code 204.	0.25
			Data Validation: Input data is properly validated for POST and PUT (0.25)	0.25	@Consumes	0.25	@Produces	0.25	@Consumes	0.25	@Consumes	0.25
				0.00	return data @Produces missing	0.00	Get all method and getby id if		return data @Produces missing	0.00	return data @Produces missing	0.00
			Storage Penalty : The model class need to be filled and put in a arraylist or suitable variable. Student can do this in the resource class or in the DAO class. (-0.5)				no results 404 missing	-0.25	invalid item or failed with 200	0.00	invalid item or failed with 200	0.00
CustomerResource Demo * 1 mark moved to using a DAO	9	8.50	All endpoints are demonstrated clearly in Postman (5)	4.50	GET /customers			0.50	get all need to return 404 if empty list (-0.50)			
			Including various valid and invalid inputs and expected outputs/errors	1.00	GET /customers/{id}			1.00				
			Shows effective use of HTTP methods and status codes (200,201,204,404)	1.00	POST /customers			1.00				
			Covers positive test cases		PUT /customers/{id}			1.00				
			Creating a resource with valid data, getting a resource that exists	1.00								
			Covers negative test cases		DELETE /customers/{id}			1.00				
CartResource Class	5	4.25	Includes all HTTP requests and methods that are in-line with CRUD operations. Appropriate endpoints are used for each method.(1x4).	3.50	Create		Read		Update		Delete	
			Logs provide detailed information suitable for debugging (0.25).	0.00	@POST	0.25	@GET	0.25	@PUT	0.25	@DELETE	0.25
					Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25
			Implementing and using CartNotFoundException (0.25).	0.25								
			Implementing and using OutOfStockException (0.25).	0.25	HTTP status code 201.	0.25	code 200 if data	0.25	HTTP status code 204/ 200 if data	0.25	HTTP status code 204.	0.00
			Data Validation: Input data is properly validated for POST and PUT (0.25)	0.25	@Consumes	0.25	@Produces	0.25	@Consumes	0.25	@Consumes	0.25
				0.00	return data @Produces missing	0.00	Get all method and getby id if		return data @Produces missing	0.00	return data @Produces missing	0.00
			Storage Penalty : The model class need to be filled and put in a arraylist or suitable variable. Student can do this in the resource class or in the DAO class. (-0.5)									
			Stock Management Penalty: Correctly handles stock checks and updates when adding/removing cart items. (-0.5)	0.00			no results 404 missing	-0.25	invalid item or failed with 200	0.00	invalid item or failed with 200	0.00

CartResource Demo * 1 mark moved to using a DAO	9	8.00	All endpoints are demonstrated clearly in Postman (4)	3.00	GET /customers/{customerId}/cart					0.50	204 expected	
			Including various valid and invalid inputs and expected outputs/errors	1.00	POST /customers/{customerId}/cart/items					1.00		
			Shows effective use of HTTP methods and status codes (200,201,204,404)	1.00	PUT /customers/{customerId}/cart/items/{bookId}					1.00		
			Covers positive test cases - Adding a book , getting a cart for a given customer id	1.00	DELETE /customers/{customerId}/cart/items/{bookId}					0.50		
			Covers negative test cases - Adding a book with no stock, removing a book that was not added	1.00	get all need to return 404 if empty list (-0.50)							
			No crashes during the demo	1.00								
OrderResource Class	5	3.50	Includes all HTTP requests and methods that are in-line with CRUD operations. Appropriate endpoints are used for each method.(1x4).	2.75	Create		Read		Update		Delete	
			Logs provide detailed information suitable for debugging (0.25).	0.00	@POST	0.25	@GET	0.25	@PUT	0.00	@DELETE	0.00
			Implementing and using CartNotFoundException (0.25).	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25	Media Type to be specified.	0.25
			Implementing and using OutOfStockException (0.25).	0.25	HTTP status code 201.	0.25	code 200 if data	0.25	HTTP status code 204/ 200 if data	0.00	HTTP status code 204.	0.00
			Data Validation: Input data is properly validated for POST and PUT (0.25)	0.25	@Consumes	0.25	@Produces	0.25	@Consumes	0.25	@Consumes	0.25
			Storage Penalty : The model class need to be filled and put in a arraylist or suitable variable. Student can do this in the resource class or in the DAO class. (-0.5)	0.00	return data @Produces missing	0.00	Get all method and getby id if		return data @Produces missing	0.00	return data @Produces missing	0.00
			Stock Management Penalty: Correctly updates the final stock after creating order. (-0.5)	0.00			no results 404 missing	-0.25	invalid item or failed with 200	0.00	invalid item or failed with 200	0.00
OrderResource Demo * 1 mark moved to using a DAO	9	6.25	All endpoints are demonstrated clearly in Postman (5)	2.50	GET /customers/{customerId}/orders					0.50	Could have implemented PUT and DELETE. get all need to return 404 if empty list (-0.50)	
			Including various valid and invalid inputs and expected outputs/errors	1.00	GET /customers/{customerId}/orders/{orderId}					1.00		
			Shows effective use of HTTP methods and status codes (200,201,204,404)	0.75	POST /customers/{customerId}/orders					1.00		
			Covers positive test cases - Create order with valid cart for a given customer id	1.00	PUT /customers/{customerId}/orders/{orderId}					0.00		
			Covers negative test cases - Create order with no cart. Create order with a book in the cart with insufficient stock	1.00	DELETE /customers/{customerId}/cart/items/{bookId}					0.00		
	70	60.25										
Implementing Exception Mapper												
BookNotFoundException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
AuthorNotFoundException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
CustomerNotFoundException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
OutOfStockException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
CartNotFoundException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
InvalidInputException	1	1	Exception Mapper to catch the specific exception. if global mapper was defined 0.25									
web.xml configuration or code level config	1	1	Correct configuration and application path has been set up and exceptionmapper class has been correctly included in the web.xml file									

	7	7	
Code Modularity and Organization			
Code Organization	1	1	Code is well-organized and follows a logical structure, indentation (-1)
Source code comments * * some would be AI generated comments.	1	0.5	Code is commented, providing clear explanations for complex logic or algorithms. If each and every line is commented it is an indication of AI generated comments (-1)
	2	1.5	
Documentation			
The report has a brief introduction	1	1	
Postman Collection	1	0.5	postman collection file is attached. not all tests shown on video are included, such as edge case tests
Test Plan	5	3	comprehensive test cases (valid, invalid (-1), edge, error (-1)) with all specified details (endpoint, method, description, request/response bodies, expected status, actual result), DELETE report states 200 for cart with PASS where as other tests shows 204
Report Layout and formatting	1	0.5	is well-organized and clearly written
	8	5	